

NIRF Rankings: Trends, Variations, and Insights

MTH208 COURSE PROJECT



Background



The National Institutional Ranking Framework (NIRF) is an initiative by the Ministry of Education, Government of India. It was launched in 2015 to rank higher education institutions based on different performance parameters. NIRF publishes rankings annually in a number of categories, including Overall, Engineering, Management, University, and College. The ranking parameters are:

- Teaching, Learning, and Resources (TLR) (30%)
- Research and Professional Practice (RPC) (30%)
- Graduation Outcomes (GO) (20%)
- Outreach and Inclusivity (OI) (10%)
- Perception (PR) (10%)

We are interested in analyzing the relative influence of each metric and the evolution of institutional performance over time. This project aims to build an interactive R Shiny web application for visualizing and analyzing NIRF data (2017–2025) through statistical methodology and trends.

Motivation

- The National Institutional Ranking Framework (NIRF) has emerged as one of the **most influential tools** for evaluating higher education institutions in India.
- In recent years, students have increasingly relied on NIRF rankings to make informed decisions about **college selection**, highlighting the framework's growing significance in shaping **academic and career choices**.
- This project was initiated to move beyond the rank lists and conduct a comprehensive analysis (2017–2025) of NIRF data.
- The goal is to identify trends and patterns within the Indian higher education landscape, and to better understand how the ranking system reflects institutional excellence and development.

Workflow



Web Scrapping

Collecting and structuring
NIRF data from official
website



Shiny App Development

Creating an interactive
platform for data exploration



Analysis & Insights

Uncovering trends and
patterns in India's higher
education rankings

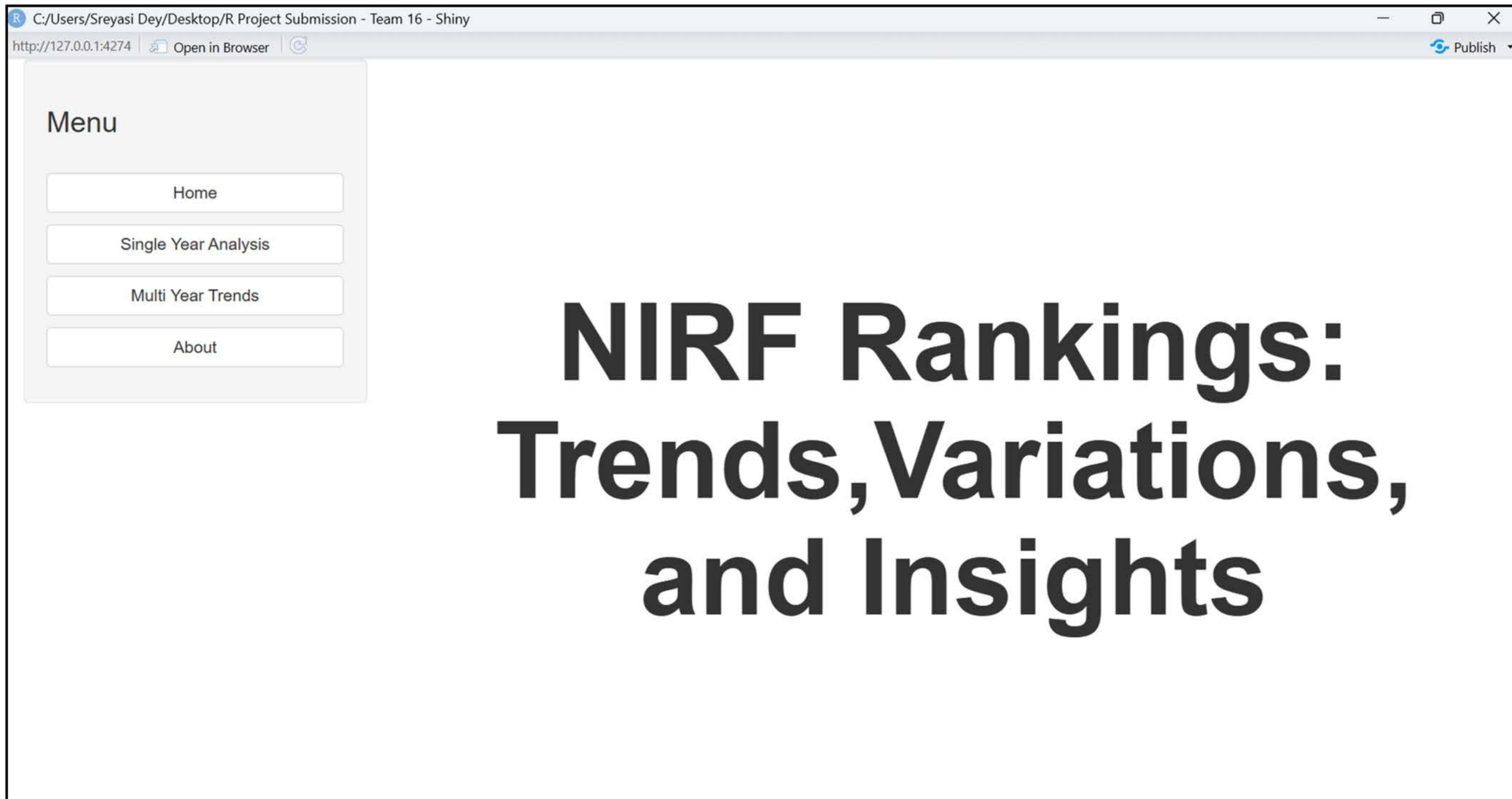


Data Collection via Web Scrapping

▶ From the official NIRF website, we have extracted the datasets for each year from 2017-2025 via web scrapping using the **rvest** and **dplyr** packages in R.

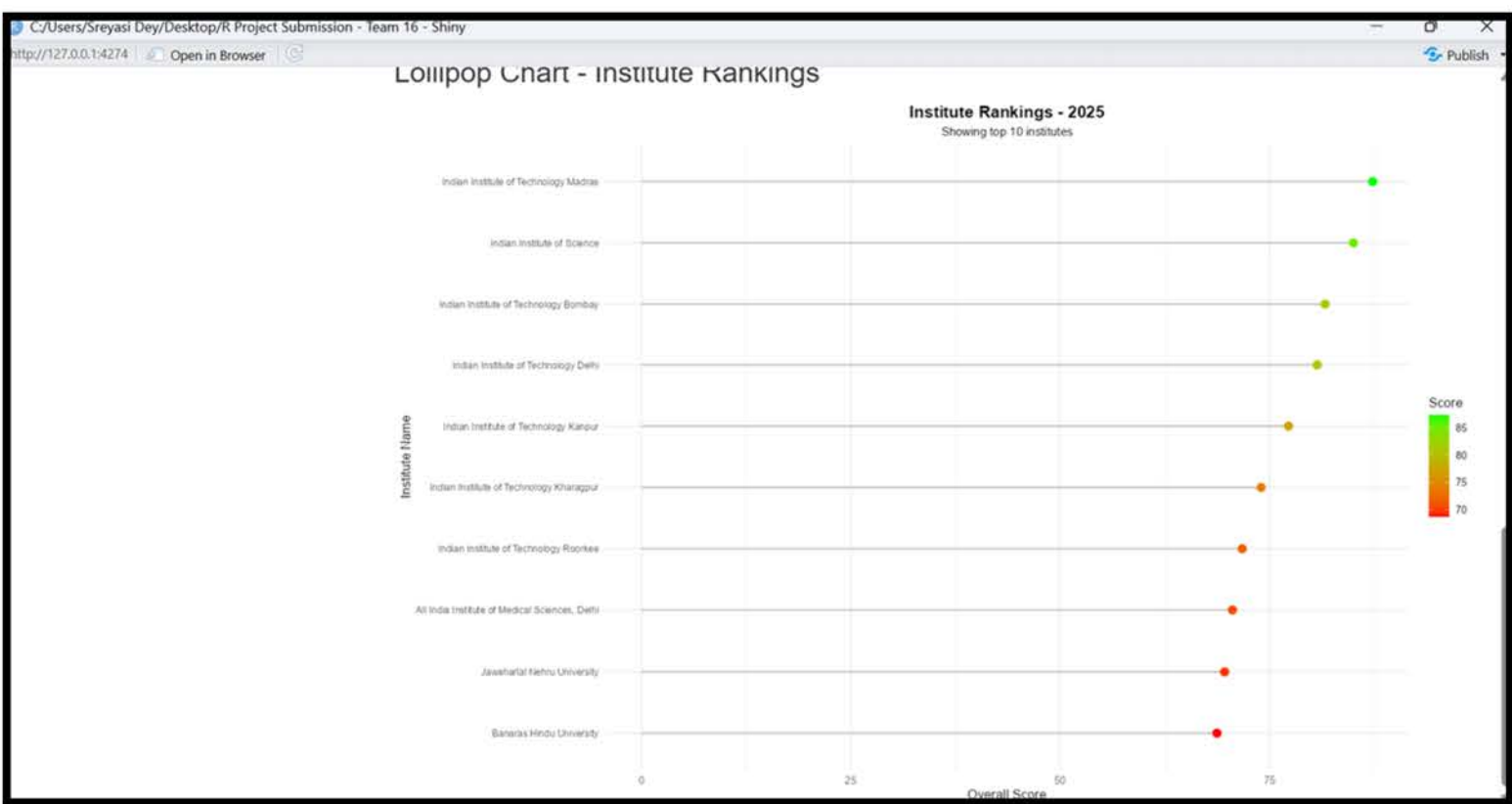
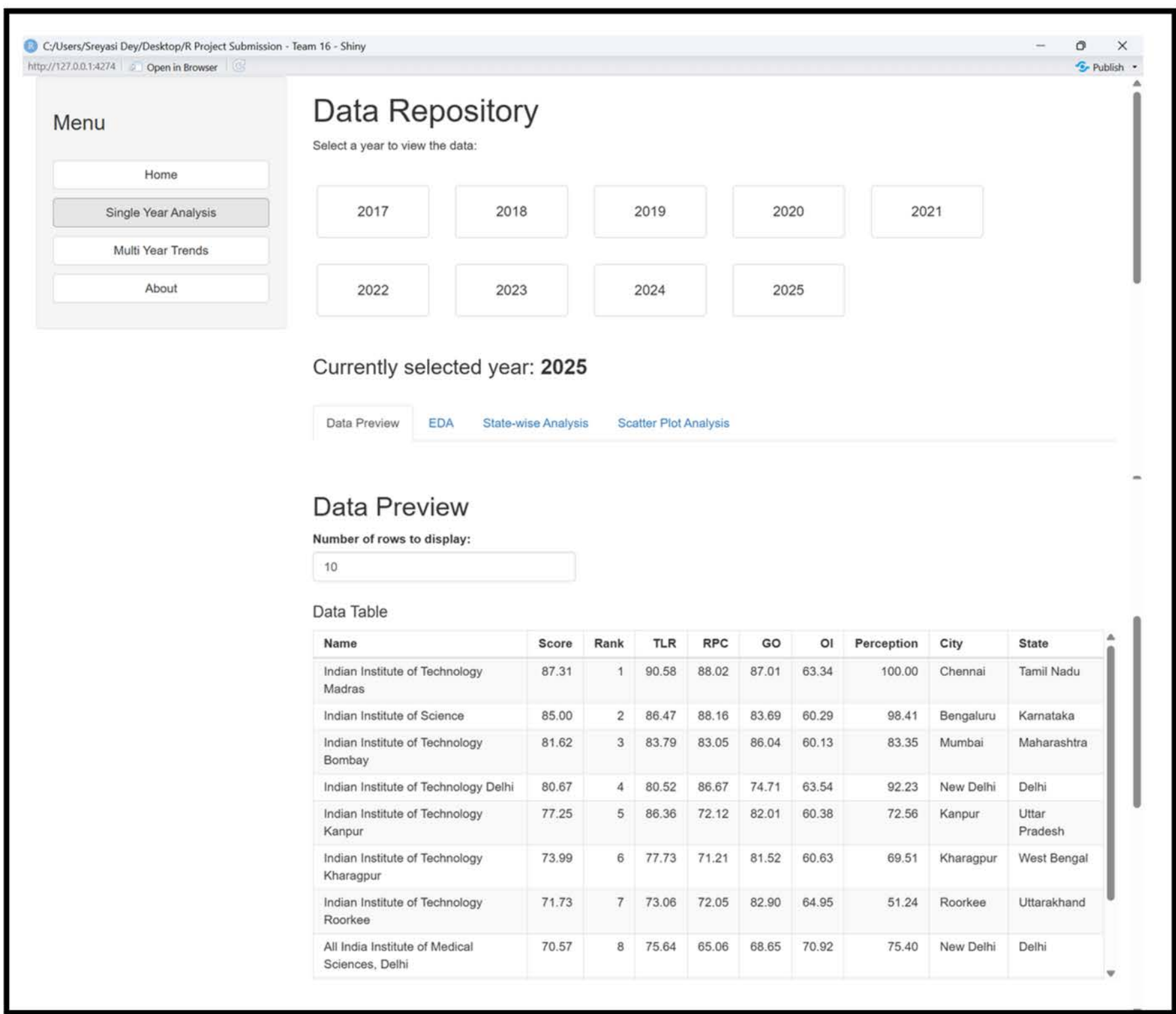
▶ All the 9 datasets have been properly cleaned to ensure consistent column names and formats across files.

Shiny App Development



A clean Home Page displaying the title and the menu to navigate in the app

Single Year Analysis



- **Users can select any year from 2017–2025.**
- **Data Table Preview:** Preview of all institutes ranks.
- **Lollipop Chart:** Institutes visualized by overall score.

Single Year Analysis

EXPLORATORY DATA ANALYSIS

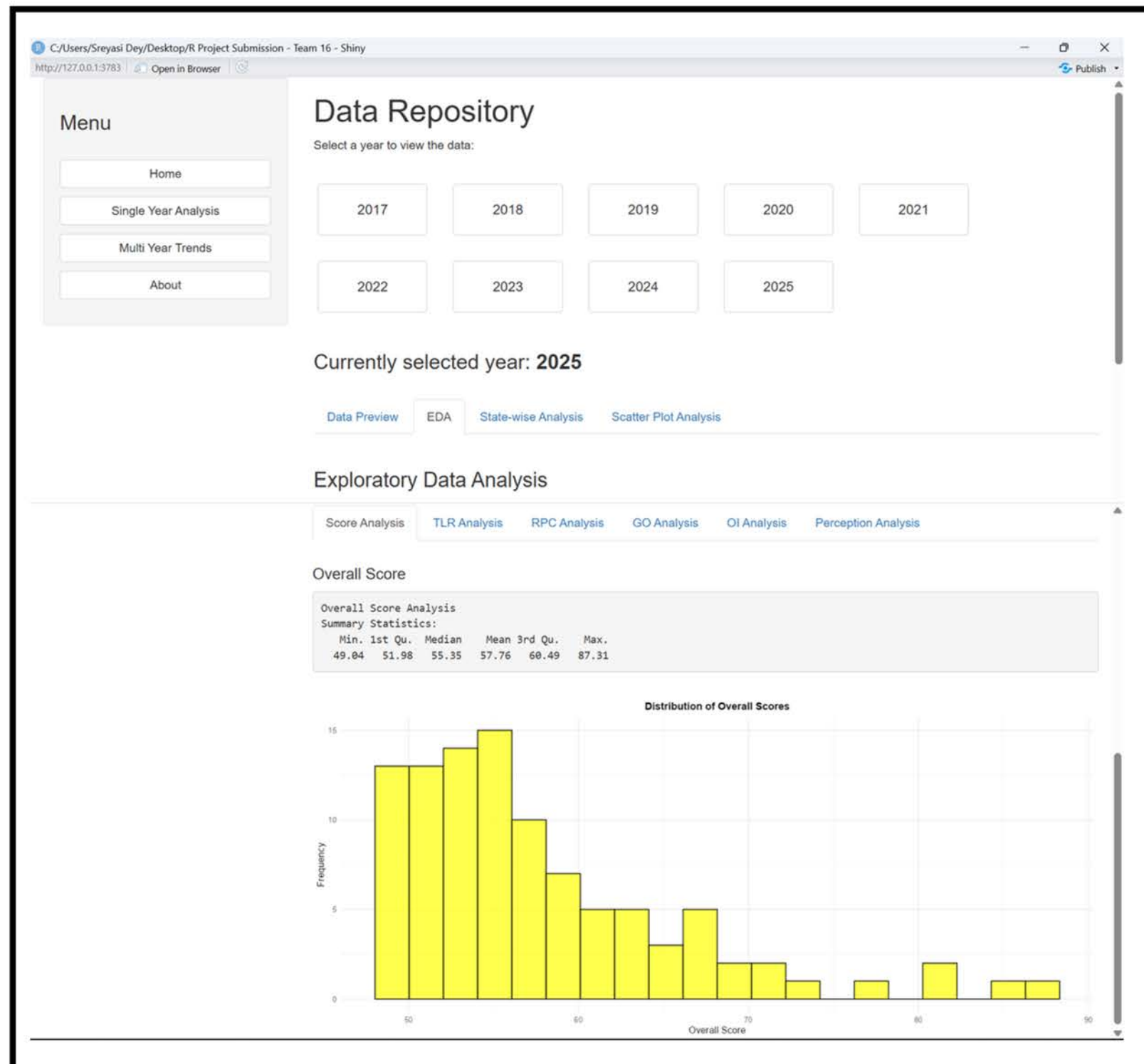
- **Summary Statistics: Mean, median, and dispersion of all the NIRF metrics.**
- **Distribution Plots: Histograms of TLR, RPC, GO, OI, PR and Score**

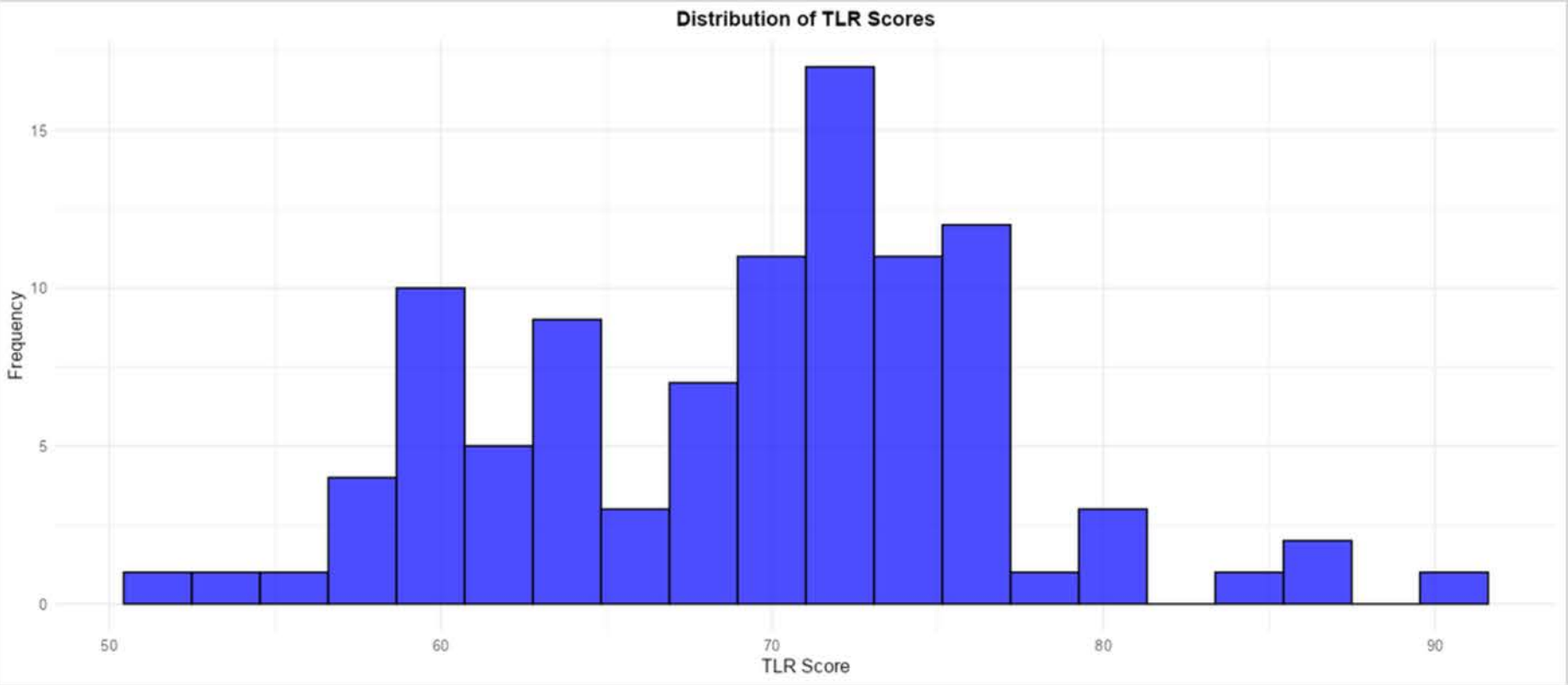
This page displays the summary statistics of overall score followed by its histogram.

The distribution of the Overall Score is skewed to the right, meaning that most institutions fall into the moderate score range (roughly 50–65), while very few attain very high overall performance.

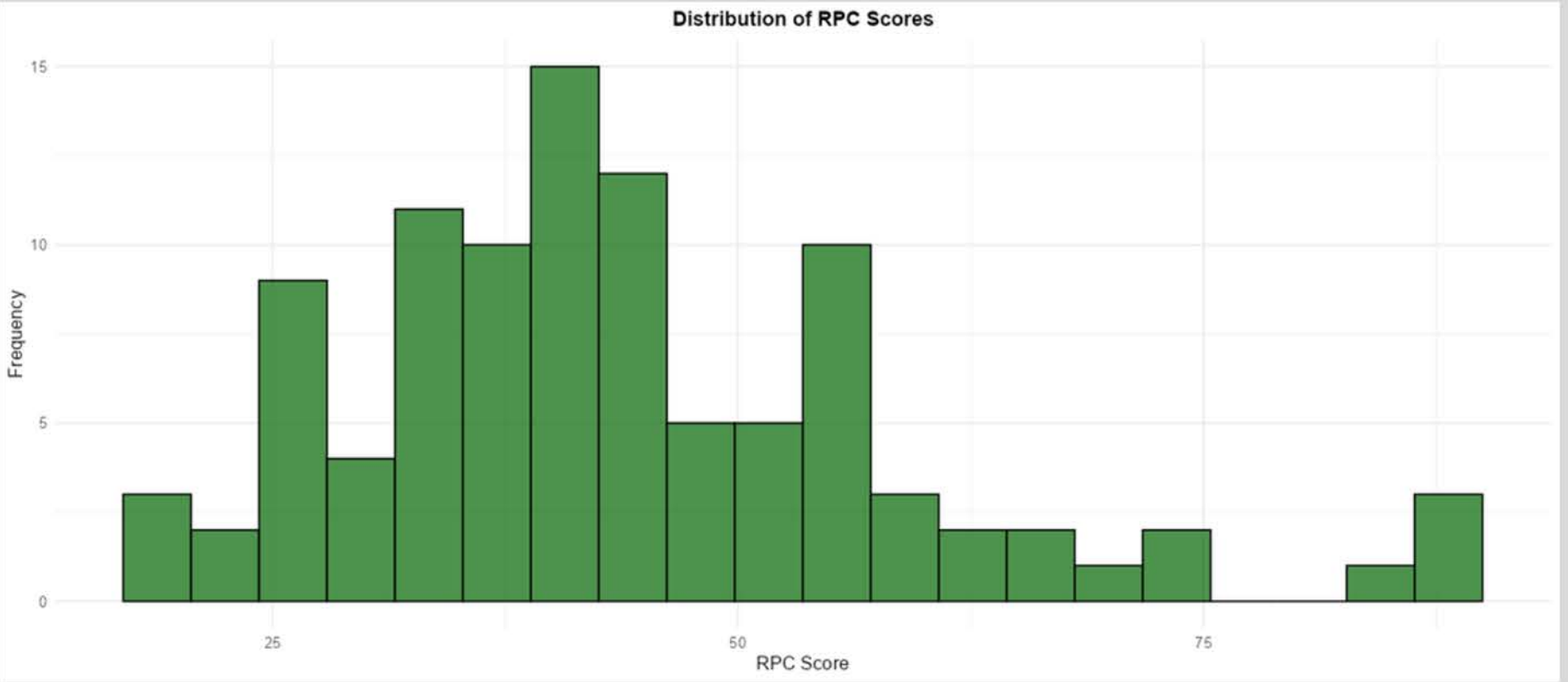
Only a few elite institutions have the ecosystem, funding, and legacy to hit very high scores.

Most others are good — but not excellent in all five NIRF dimensions, so they cluster around 40–60 marks.

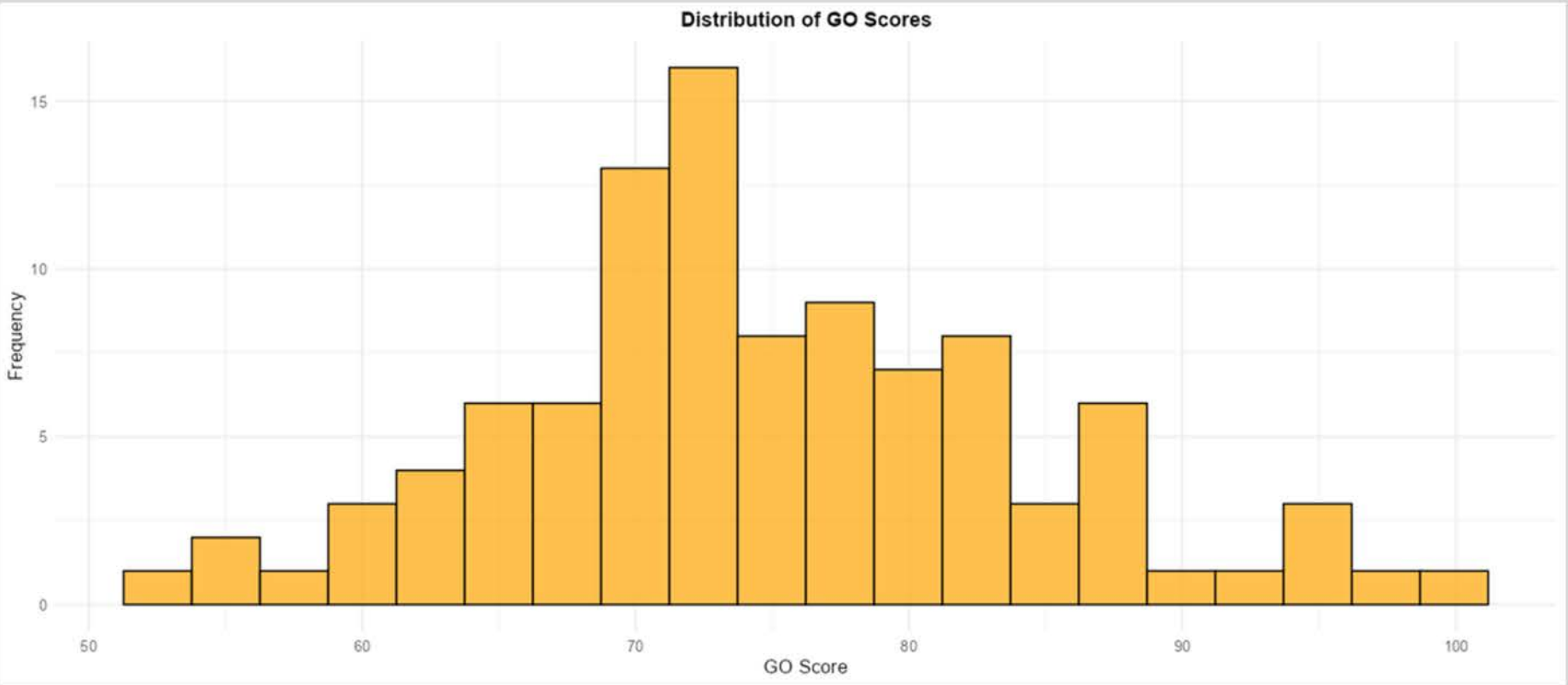




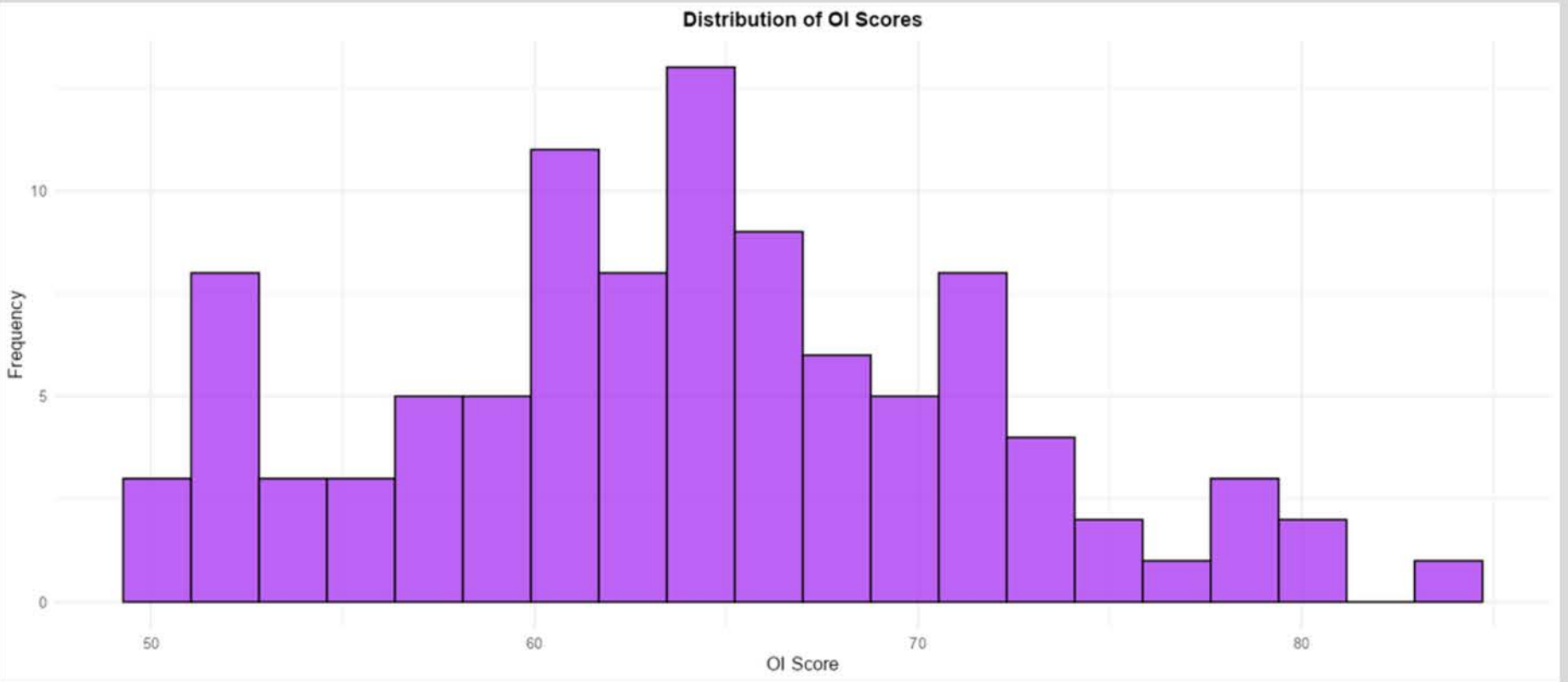
TLR: Roughly symmetric; most institutes score 60–75 — strong academic + infrastructure base.



RPC: Right-skewed; research output strong only in a few elite institutes (> 80).

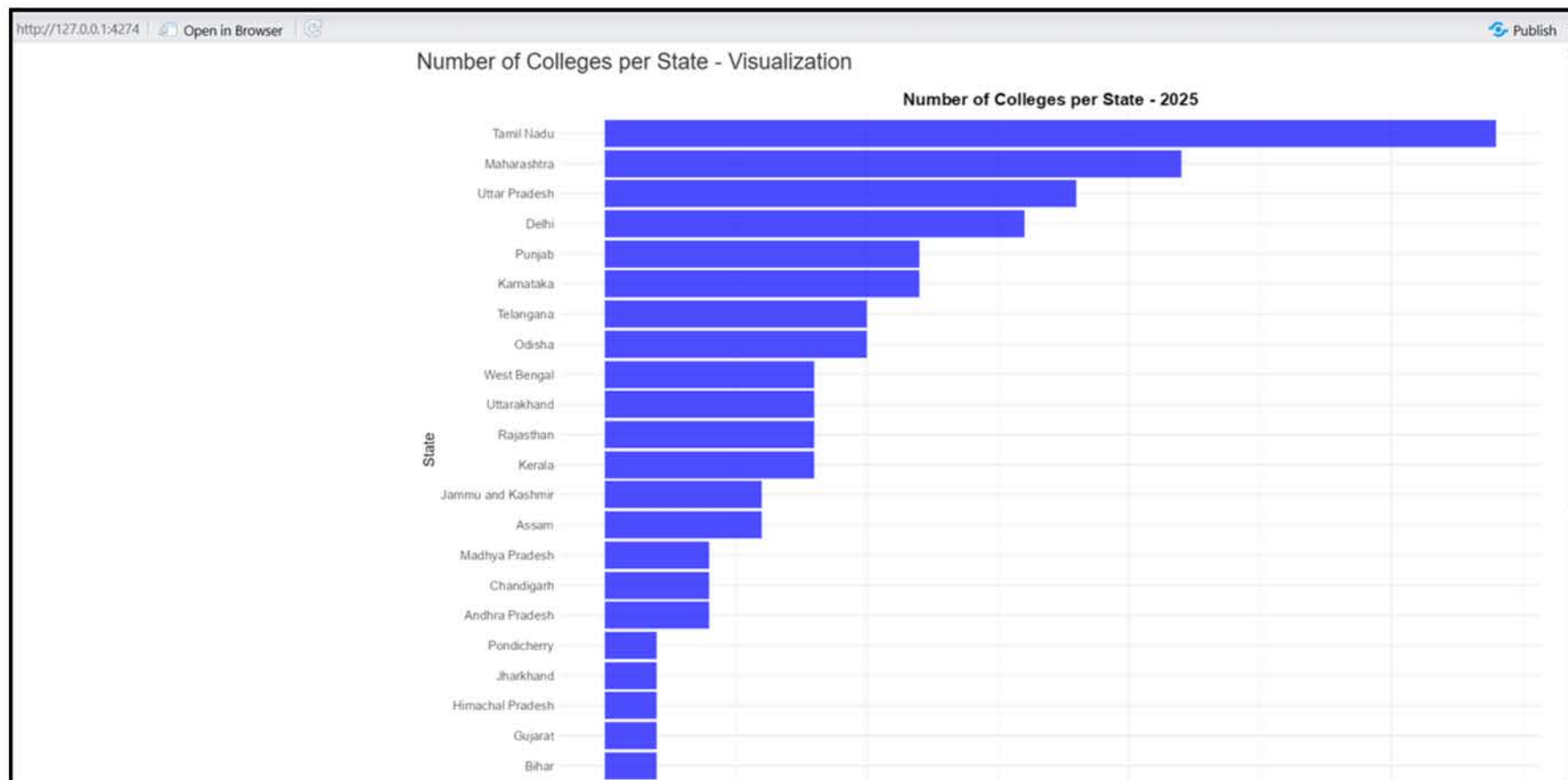
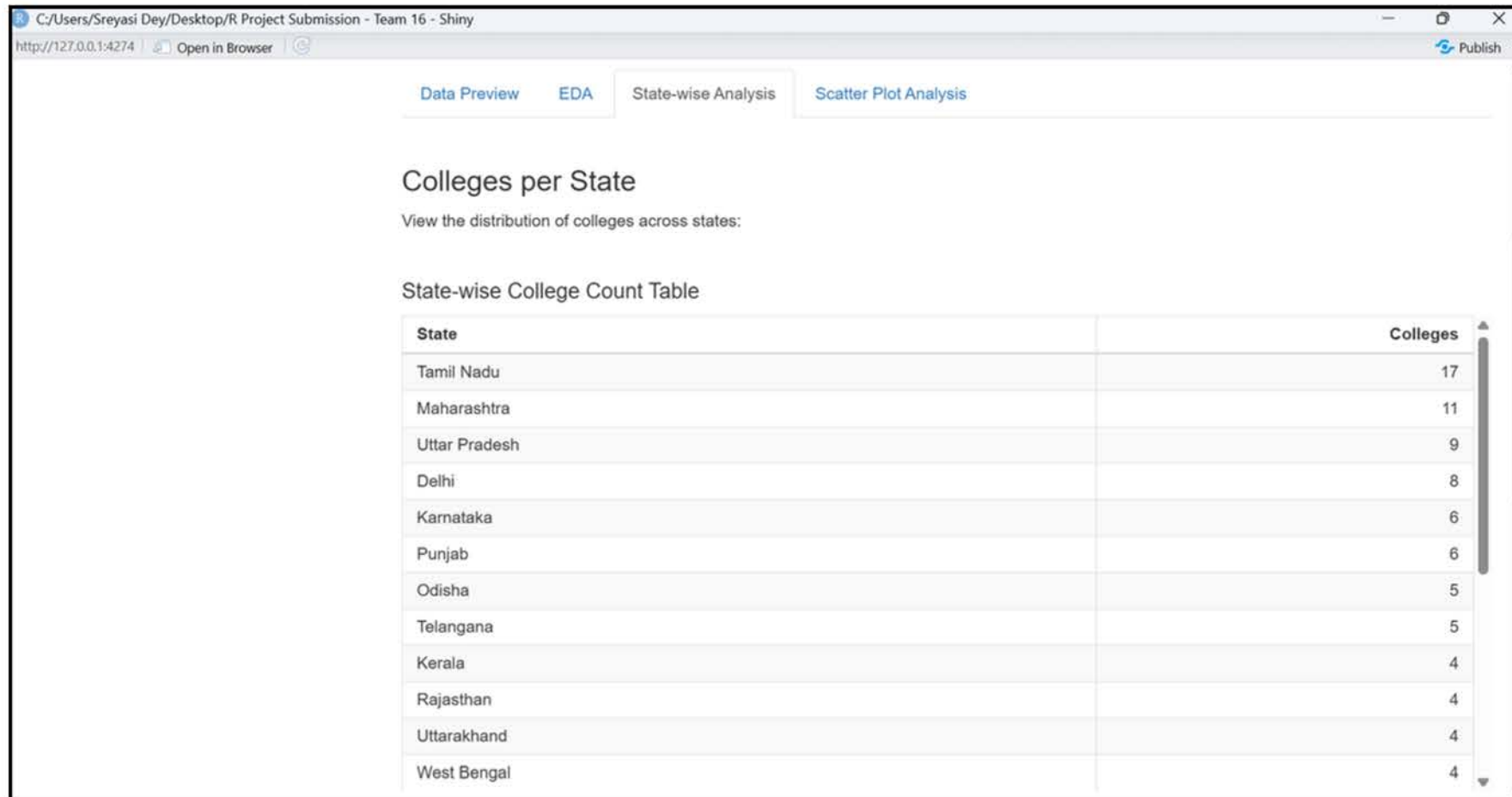


GO: Mostly 65–85; indicates good graduation + placement outcomes.



OI: Wide spread; top institutes perform well, others show scope for inclusivity.

State-wise Distribution



Regional Distribution

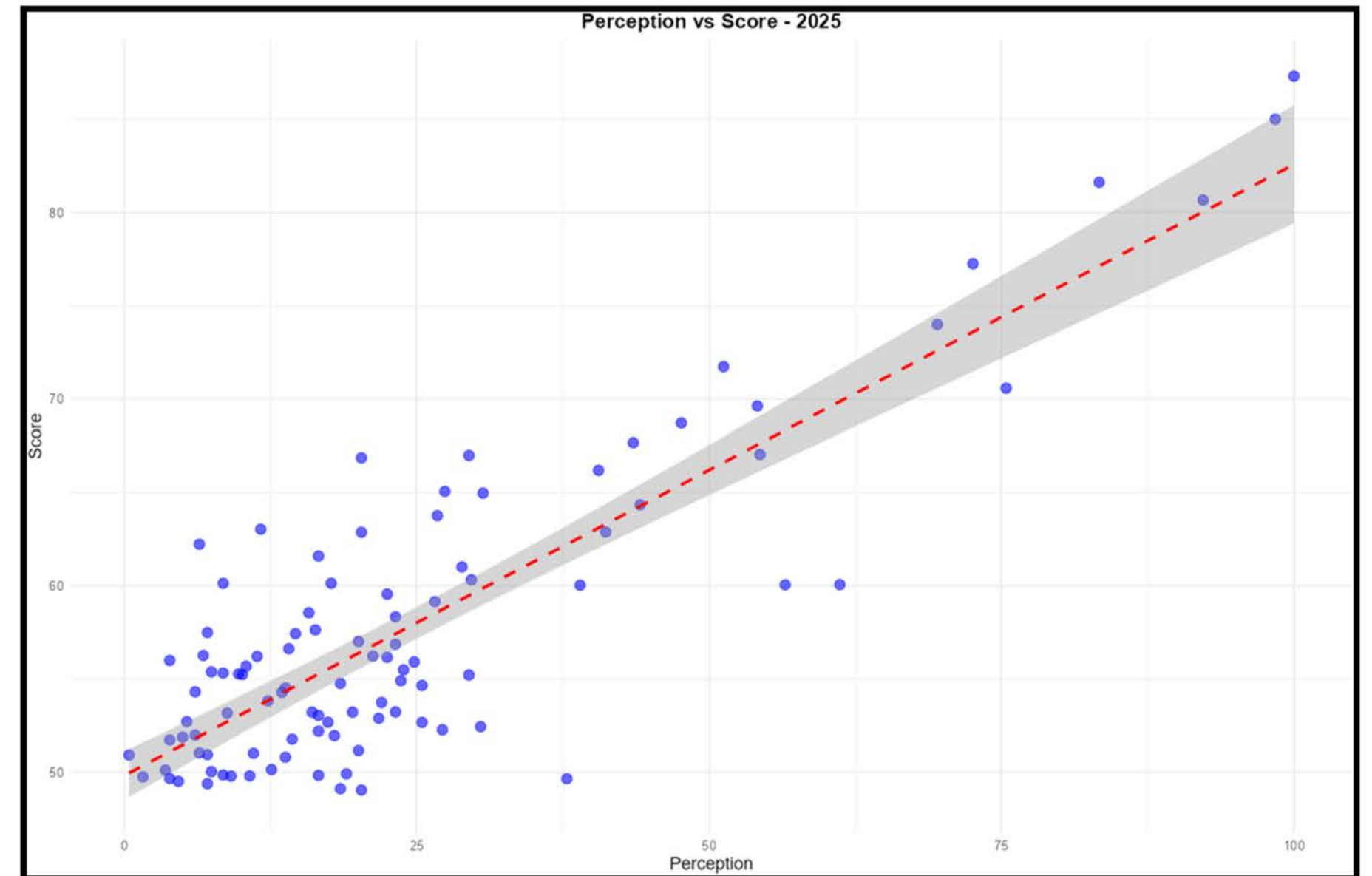
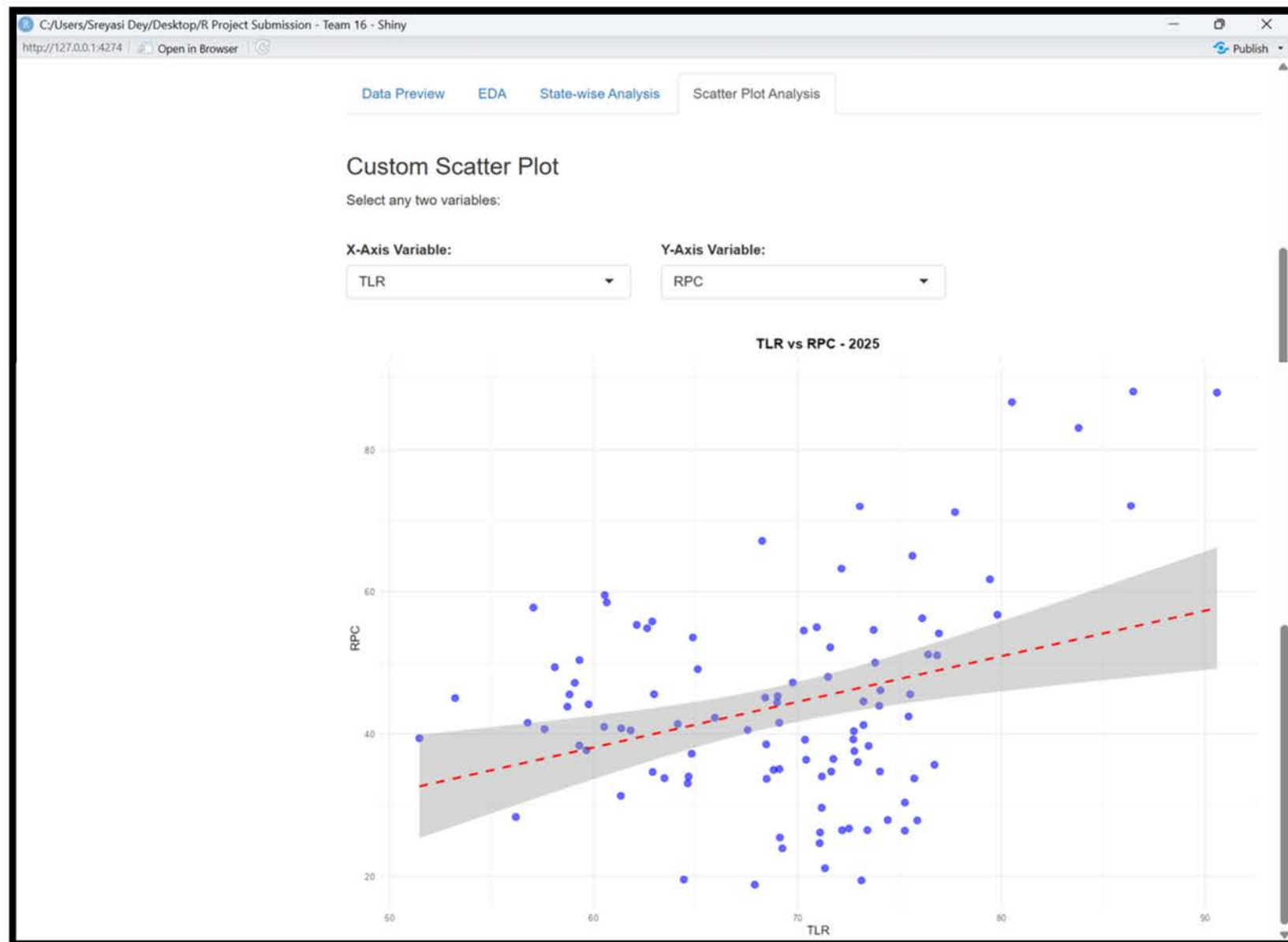
- Strong regional imbalance — Tamil Nadu, Maharashtra & Uttar Pradesh dominate the list.
- Eastern & Northeastern states show limited representation.
- Educational infrastructure remains concentrated in a few regions.

Both top states have education-friendly policies and higher budget allocations to universities, skill development, and research infrastructure.

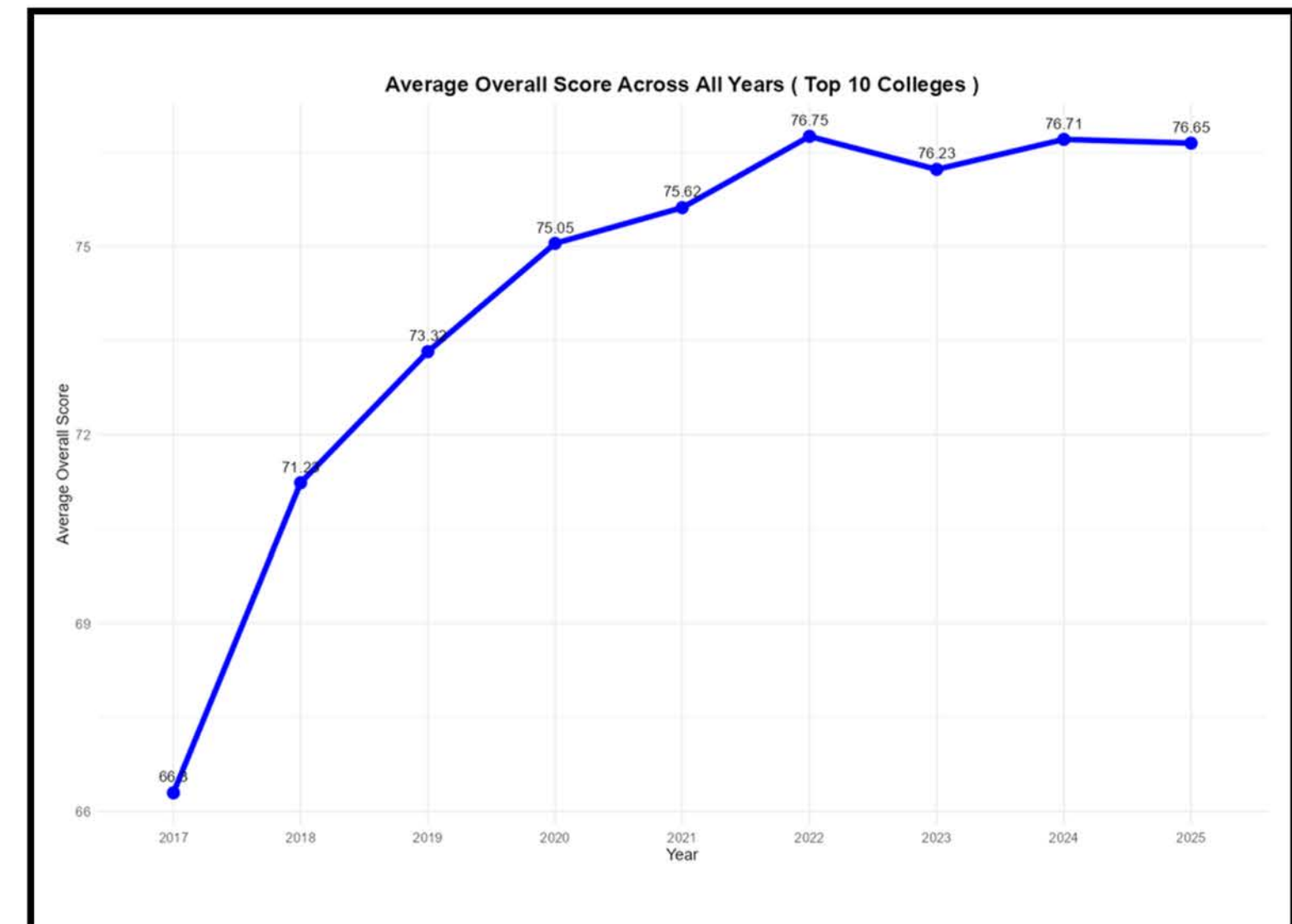
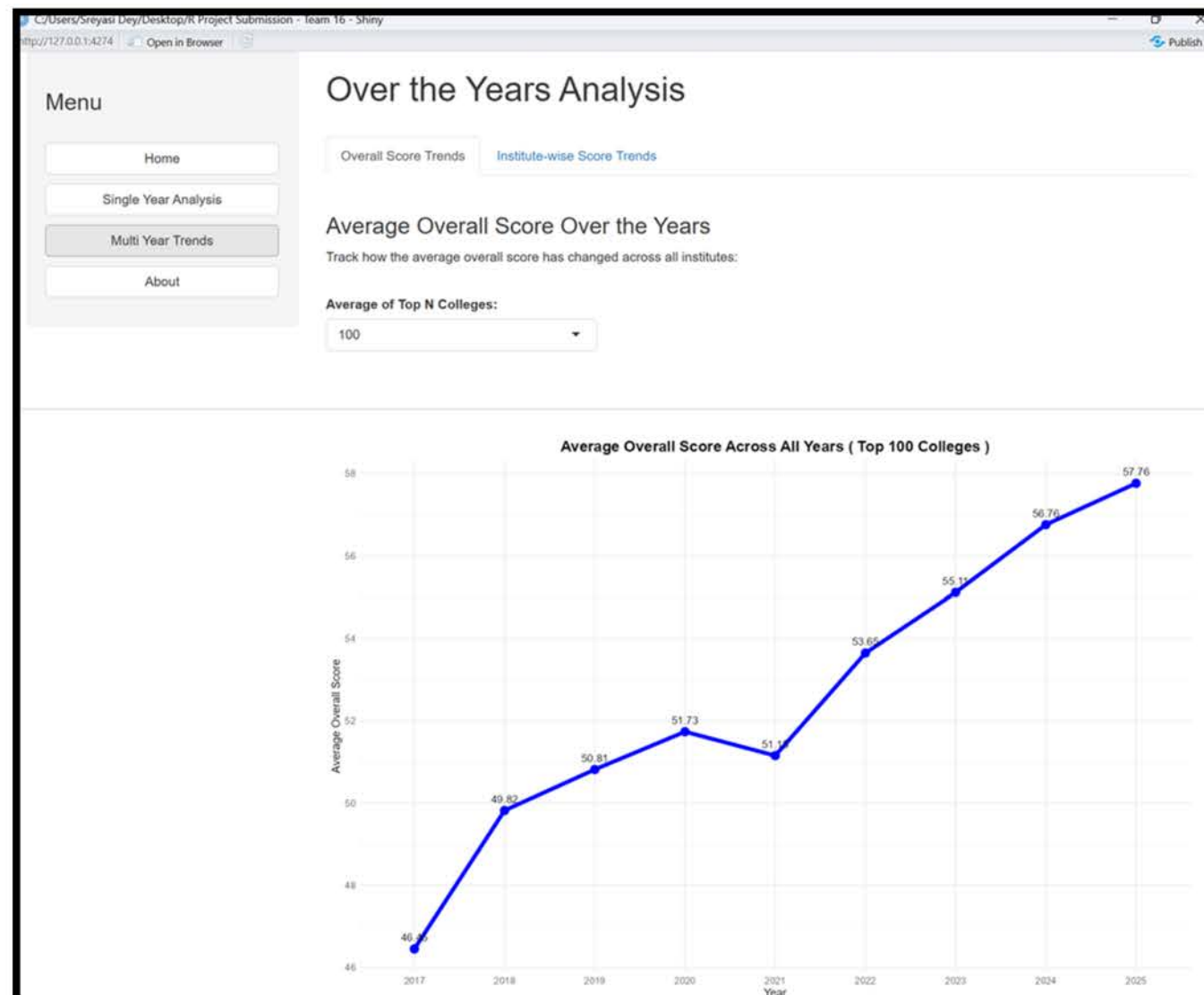
Telangana's government encourages private sector participation in education and has strong research and innovation programs (like T-Hub, WE-Hub, RICH Hyderabad).

Maharashtra's government supports autonomous universities, giving colleges flexibility to improve academic quality.

Scatterplot of any two selected parameters
(e.g., TLR vs. RPC) to study correlation.



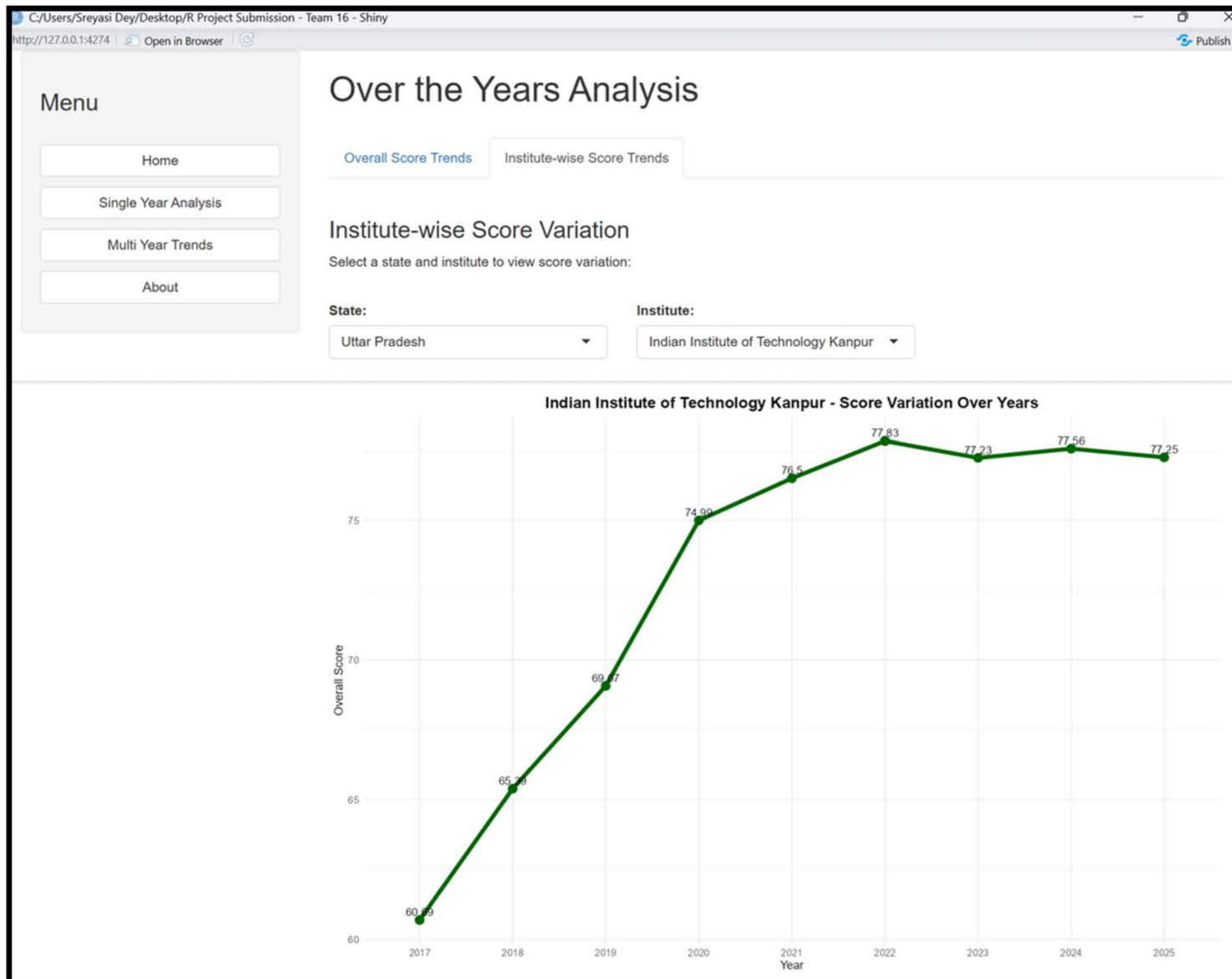
Multi-Year Trends (Overall Score Trends)



Performance Over Time

- **Upward Trend:** Avg. scores of Top 10 & Top 100 institutes improving.
- **2021 Dip:** COVID impact—larger effect on Top 100 than Top 10.
- **Post-2022 Saturation:** Top 10 scores stabilize around 77.
- **System-wide Improvement:** Mid-tier institutes showing steady growth.

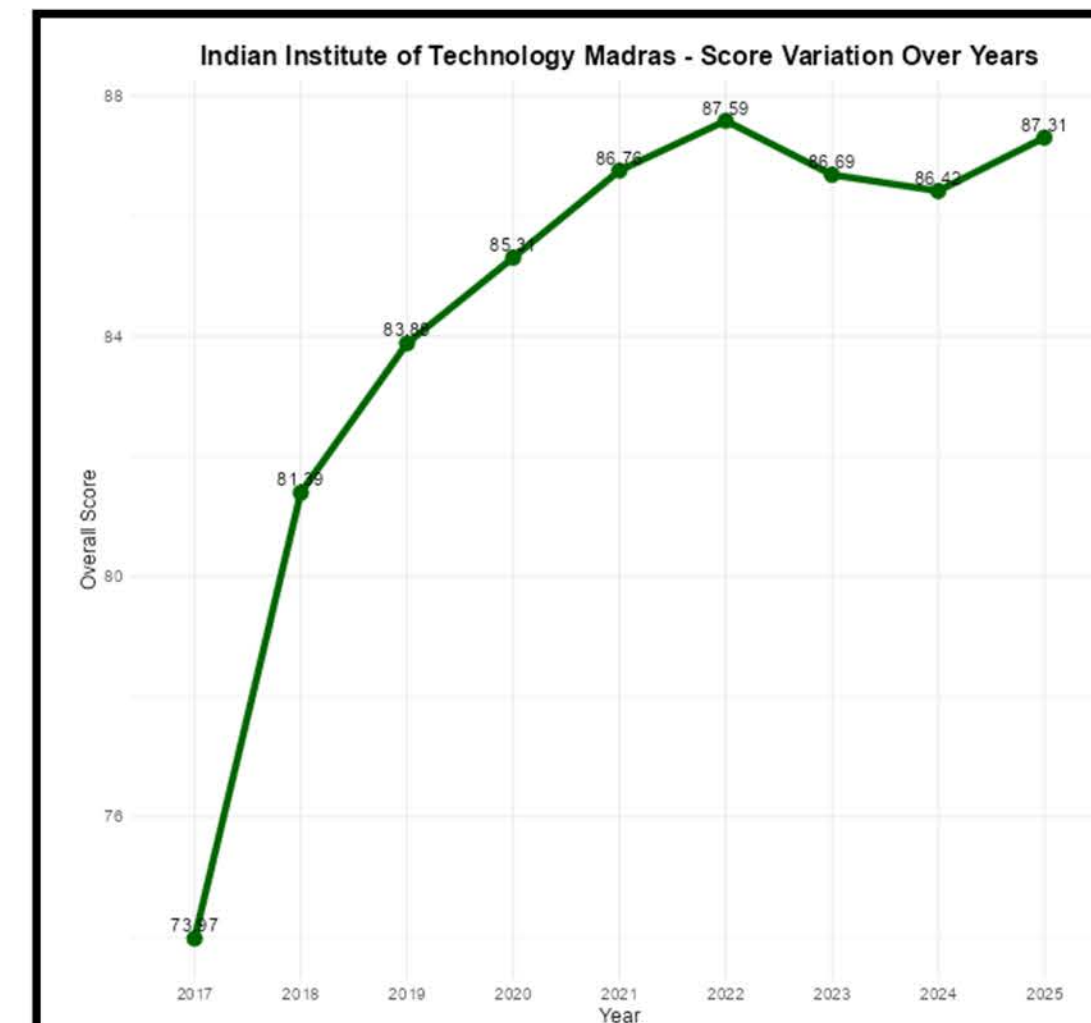
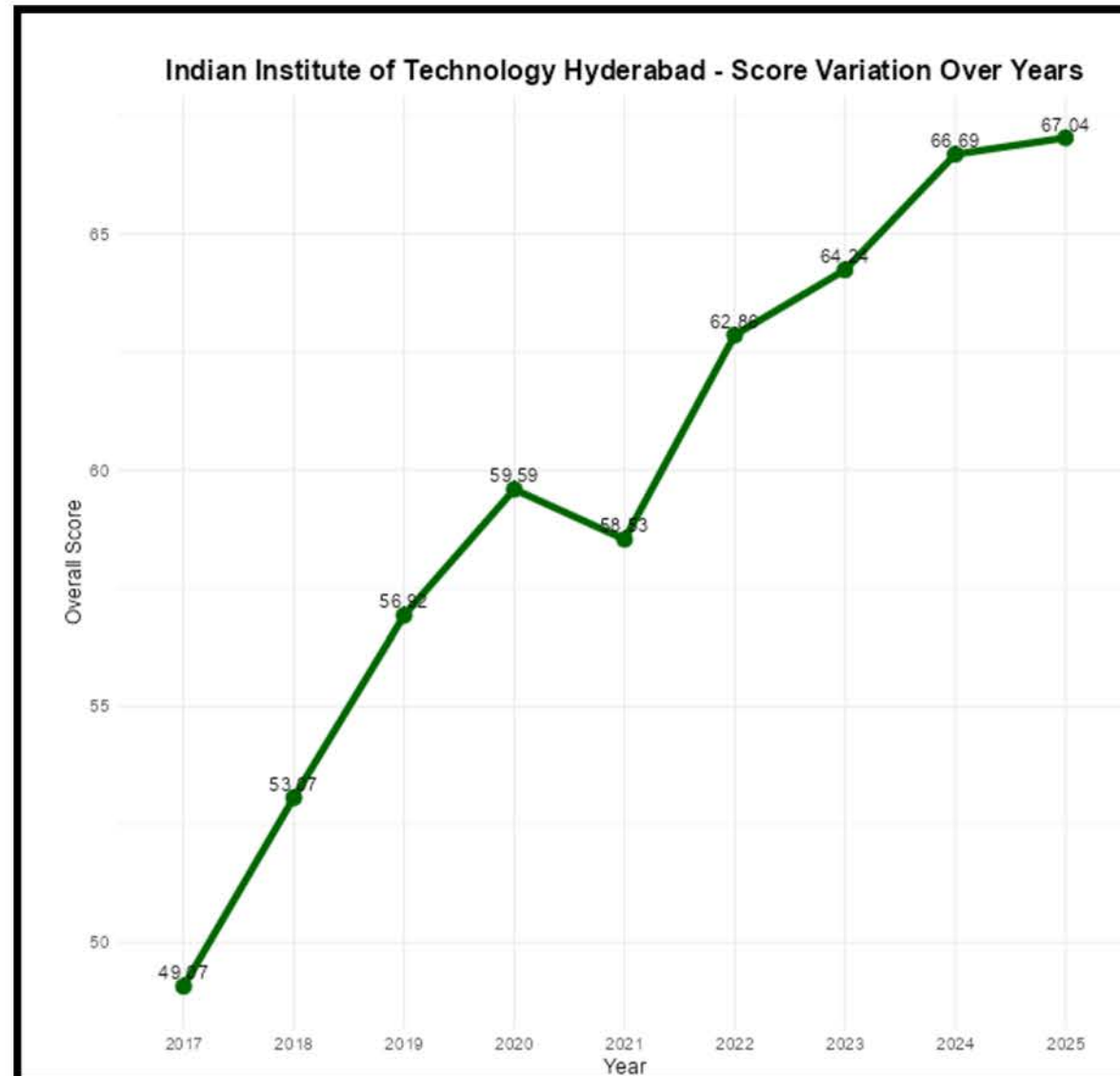
Multi-Year Trends (Institute-wise score trends)



IIT Kanpur Performance

- 2017–2021: Sharp growth (60.9 → 76.5).
- 2022 Peak: 77.83 — best performance.
- 2023–2025: Scores stable (~77–78) → consistent excellence.
- Overall trend = growth + stabilization → sustained top-tier status.

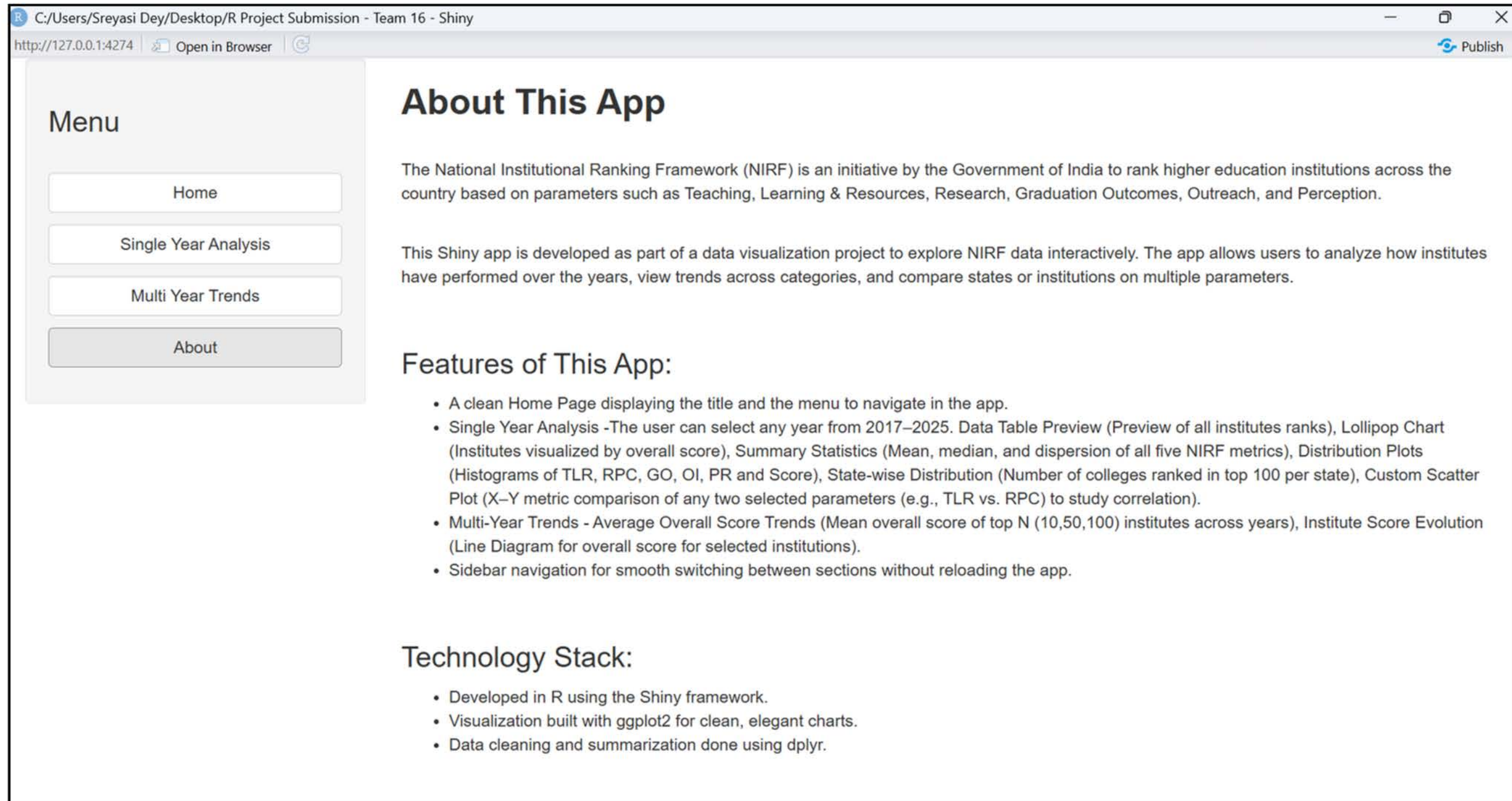
Performance between newer and older IITs



Performance Over Time

- The long-term trend shows that the IIT network is strengthening system-wide, with older institutions maintaining their excellence and newer ones catching up through innovation, research expansion, and quality enhancement. This indicates balanced growth in India's top technical education system.

About Page



Provides project
summary,
prerequisite, and
implementation
details.

Future Improvements

- Addition of filters for institution type (Engineering, University, Research, etc.).
- Addition of hypothetical ways to calculate and present Score from the same parameters.
- Implementation of dynamic dashboard and better UI.
- Implementing Interactive maps to visualize college distribution by state/city dynamically.

Conclusion

The project successfully developed an interactive R Shiny application to analyze NIRF ranking data from 2017–2025. It provides an intuitive and data-driven understanding of institutional trends, helping users visualize performance patterns and policy insights.

- Enables single-year and multi-year performance analysis
- Shows consistent dominance of IITs, IIMs, and IISc
- Highlights gradual improvement among mid-tier institutes
- Reveals regional and perception disparities
- Serves as a valuable tool for students, researchers, and policymakers

Overall, the study reinforces the importance of data-driven evaluation in strengthening India's higher education landscape.

THANK YOU